

# 2026-06-10-sdk-cli-currency

## SDK & CLI-Agent Currency Research — 2026-06-10

**Scope:** §11.4.99 latest-source deep-web research for HelixCode. Two parts: PART A = Go provider/infra SDK currency (SP2 P2.3 SDK-currency); PART B = non-interactive/scriptable invocation for bridged CLI agents (SP4 P4.3).

**Method:** Every version/flag below was fetched from the OFFICIAL source (GitHub releases, pkg.go.dev ?tab=versions, vendor docs) on 2026-06-10 — never training memory. Honest negative findings are recorded where a page was unreachable. Full URL list in the ## Sources verified 2026-06-10 footer.

**HelixCode current pins (read 2026-06-10):** - helix\_code/go.mod — aws-sdk-go-v2 v1.32.7, config v1.28.7, credentials v1.17.48, service/bedrockruntime v1.23.1, azcore v1.16.0, azidentity v1.8.0, gin v1.11.0, smacker/go-tree-sitter v0.0.0-20240827094217-dd81d9e9be82, getzep/zep-go/v3 v3.10.0. - submodules/helix\_agent/go.mod — gin v1.12.0, smacker/go-tree-sitter v0.0.0-20240827094217-dd81d9e9be82. - (helix\_agent/go.mod at repo root does not exist; the submodule lives at submodules/helix\_agent/.) - No vendor SDK for OpenAI/Anthropic/Gemini is pinned — HelixCode hand-rolls HTTP to these providers (no openai/openai-go, anthropics/anthropic-sdk-go, or google.golang.org/genai in either go.mod).

### PART A — Go provider/infra SDK latest versions

| Dependency                                       | HelixCode pin | LATEST (2026-06-10) | Released   | Skew             |
|--------------------------------------------------|---------------|---------------------|------------|------------------|
| github.com/aws/aws-sdk-go-v2 (core)              | v1.32.7       | v1.42.0             | 2026-06-08 | 10 minor behind  |
| .../service/bedrockruntime                       | v1.23.1       | v1.53.5             | 2026-06-08 | ~30 minor behind |
| github.com/Azure/azure-sdk-for-go/sdk/azcore     | v1.16.0       | v1.22.0             | 2026-06-04 | 6 minor behind   |
| github.com/Azure/azure-sdk-for-go/sdk/azidentity | v1.8.0        | v1.13.1 (stable)    | 2025-11-10 | 5 minor behind   |

| Dependency                             | HelixCode pin                                          | LATEST<br>(2026-06-10)                                                      | Released   | Skew                                                                 |
|----------------------------------------|--------------------------------------------------------|-----------------------------------------------------------------------------|------------|----------------------------------------------------------------------|
|                                        |                                                        |                                                                             |            |                                                                      |
| github.com/gin-gonic/gin               | v1.11.0 (helix_code) / v1.12.0 (helix_agent)           | <b>v1.12.0</b>                                                              | 2026-02-28 | helix_code 1 minor behind; helix_agent already current               |
| github.com/smacker/go-tree-sitter      | v0.0.0-20240827094217-dd81d9e9be82 (2024-08-27 commit) | <b>same commit is still HEAD</b> (latest master commit dd81d9e, 2024-08-27) | 2024-08-27 | At HEAD, but repo is effectively dormant (no commits since Aug-2024) |
| github.com/getzep/zep-go/v3            | v3.10.0                                                | <b>v3.23.0</b>                                                              | 2026-06-04 | 13 minor behind                                                      |
| github.com/openai/openai-go            | not used (hand-rolled HTTP)                            | <b>v1.12.0</b>                                                              | 2025-07-30 | n/a                                                                  |
| github.com/anthropics/anthropic-sdk-go | not used (hand-rolled HTTP)                            | <b>v1.50.1</b>                                                              | 2026-06-09 | n/a                                                                  |
| google.golang.org/genai (Gemini)       | not used (hand-rolled HTTP)                            | <b>v1.59.0</b>                                                              | 2026-06-03 | n/a                                                                  |

**go-tree-sitter migration note (negative/structural finding)**

smacker/go-tree-sitter is pinned at its HEAD commit but the repo has had **no commits since 2024-08-27** (verified on the commits/master page). The OFFICIAL binding github.com/tree-sitter/go-tree-sitter (in the upstream tree-sitter org) is actively released — latest tag **v0.25.0 (2025-02-02)**. The official README does NOT contain explicit “supersedes smacker/...” language (honest negative finding — I did not find a vendor statement of supersession), so the migration is a judgment call, not a vendor-mandated one. If HelixCode wants a maintained binding it would move smacker → tree-sitter/go-tree-sitter v0.25.0; note that is a different import path + API (Close-on-allocation requirement), so it is real migration work, not a version bump.

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## PART B — Non-interactive / scriptable invocation for bridge CLI agents

Confirmed CURRENT (2026-06-10) command forms. “JSON output” column = does the agent emit machine-readable JSON in headless mode today.

| CLI agent                                         | Single non-interactive prompt                                                                              | JSON / machine-readable output                                                                                                                         | List models                                                                                                        | Auth / env                                                                                                                                                                                                                          |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Claude Code</b><br>(claude)                    | <code>claude -p "&lt;prompt&gt;"</code><br>(--print/-p)                                                    | <b>Yes</b> — --output-format json (also text, stream-json); --input-format text\ stream-json; --json-schema '<schema>' for validated structured output | No dedicated models list command; set with --model sonnet\ opus\ haiku\ fable\ <full-name>; --fallback-model chain | ANTHROPIC_API_KEY or Claude subscription; claude setup-token mints a long-lived OAuth token for CI                                                                                                                                  |
| <b>Qwen Code</b><br>(qwen)                        | <code>qwen -p "&lt;prompt&gt;"</code><br>(-p/--prompt, “headless mode”)                                    | <b>Yes</b> — --output-format text\ json\ stream-json; --input-format text\ stream-json; --include-partial-messages                                     | <b>No</b> CLI flag to list models (config-file / interactive / model only)                                         | API-key / OAuth via qwen auth subcommands (qwen auth api-key, qwen auth status); -y/--yolo auto-approve                                                                                                                             |
| <b>OpenCode</b><br>(opencode)<br>incl. <b>Zen</b> | <code>opencode run "&lt;prompt&gt;"</code><br>(args or stdin); --attach http://host:port to reuse a server | <b>Yes</b> — opencode run --format json (raw JSON events; default = formatted)                                                                         | <b>Yes</b> — opencode models [provider] (--refresh, --verbose); for Zen also GET https://opencode.ai/zen/v1/models | Zen: sign in to OpenCode Zen, copy API key, /connect → OpenCode Zen → paste key. <b>Zen model id format</b> = opencode/<model-id> (e.g. opencode/gpt-5.5); Zen is a curated 40+ model gateway (GPT/Claude/Gemini/Qwen/DeepSeek/...) |
| <b>Gemini CLI</b><br>(gemini)                     | <code>gemini -p "&lt;prompt&gt;"</code><br>(-p/--prompt; headless also auto-triggers in non-TTY)           | <b>Yes</b> — --output-format text\ json; JSON is a single object with response (string), stats                                                         | No documented list-models flag (honest negative finding); set with -m/--model                                      | Gemini API key / Google auth env; standard Gemini CLI config                                                                                                                                                                        |

| CLI agent                              | Single non-interactive prompt                                                                                              | JSON / machine-readable output                                                                                                                                | List models                                                                                                                                                                                                                  | Auth / env                                                                                                          |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
|                                        |                                                                                                                            | (token/latency), error (optional)                                                                                                                             |                                                                                                                                                                                                                              |                                                                                                                     |
| <b>Crush</b><br>(crush)                | crush run " <code>&lt;prompt&gt;</code> "<br>(args or piped stdin; single-turn → stdout)                                   | <b>No</b> dedicated JSON output format found (honest negative finding — README + run docs show no --format json); --quiet suppresses spinner for clean stdout | Models via config file ( <code>.crush.json</code> / <code>crush.json</code> / <code>~/.config/crush/crush.json</code> ); <code>crush update-providers</code> refreshes provider list; no <code>models</code> list flag found | Provider keys via config/env; --yolo auto-approves tools (non-interactive); global flags --debug, --cwd, --data-dir |
| <b>Codex CLI</b><br>(codex)            | codex exec " <code>&lt;prompt&gt;</code> "<br>(alias <code>codex e</code> )                                                | <b>Yes</b> — --json / --experimental-json (newline-delimited JSON events); -o/--output-last-message <code>&lt;path&gt;</code> writes final message to file    | No documented list-models flag; set with -m/--model                                                                                                                                                                          | OpenAI auth; --sandbox read-only\ workspace-write\ danger-full-access; --ephemeral for no session files             |
| <b>Goose</b><br>(goose)                | goose run -t " <code>&lt;prompt&gt;</code> "<br>(-t/--text; -i/--instructions <code>&lt;file&gt;</code>  ->; --no-session) | <b>Yes</b> — --output-format text\ json\ stream-json                                                                                                          | No CLI list-models flag (honest negative finding); set with --provider + --model                                                                                                                                             | Provider/model via env or --provider/--model flags                                                                  |
| <b>GitHub Copilot CLI</b><br>(copilot) | copilot -p " <code>&lt;prompt&gt;</code> "<br>(-p/--prompt, non-interactive)                                               | <b>No</b> JSON format flag found (honest negative finding); -s/--silent strips stats/decoration → clean text for piping                                       | <b>No</b> list flag; model strings are shown in the --model option description under copilot help; set with --model gpt-5.2\ claude-sonnet-4.6\ ...                                                                          | GitHub auth; --no-ask-user prevents clarifying-question pauses; --allow-all-tools for unattended runs               |

## 12-line summary

1. **Most-stale SDK (priority #1):** service/bedrockruntime **v1.23.1** → **v1.53.5** — ~30 minor versions behind; same v1 line, non-breaking.

2. **Priority #2:** getzep/zep-go/v3 **v3.10.0** → **v3.23.0** — 13 minor behind, same v3 line.
  3. **Priority #3:** aws-sdk-go-v2 **core v1.32.7** → **v1.42.0** (+ config/credentials siblings) — 10 minor behind; new retry behavior is opt-in behind `AWS_NEW_RETRIES_2026`.
  4. **Priority #4:** azidentity **v1.8.0** → **v1.13.1** — 5 minor behind AND carries security relevance (GO-2024-2918 EoP on old releases); stay on stable, v1.14.0 is beta-only.
  5. **Priority #5:** azcore **v1.16.0** → **v1.22.0** — 6 minor behind, non-breaking.
  6. **gin skew fix: bump helix\_code v1.11.0** → **v1.12.0** to match helix\_agent (already on v1.12.0); additive features only.
  7. **All Part-A bumps are same-major (no breaking major bump required)** for currently-pinned deps.
  8. **go-tree-sitter:** pinned at smack HEAD but that repo is dormant since Aug-2024; maintained path is `tree-sitter/go-tree-sitter` v0.25.0 — a real migration (different import path/API), not a version bump; no vendor supersession statement found.
  9. **Vendor SDKs HelixCode does NOT use (hand-rolled HTTP):** openai-go v1.12.0, anthropic-sdk-go v1.50.1, google.golang.org/genai v1.59.0 — adopting any is net-new dependency work, all on v1.x.
  10. **CLI agents with CONFIRMED non-interactive JSON output today:** Claude Code (`--output-format json`), Qwen Code (`--output-format json`), OpenCode (`run --format json`), Gemini CLI (`--output-format json`), Codex CLI (`exec --json`), Goose (`--output-format json`).
  11. **CLI agents WITHOUT a JSON output format (text-only, use -s/--quiet for clean stdout):** GitHub Copilot CLI (`-p -s`) and Crush (`run + --quiet`) — honest negative finding from their official docs.
  12. **OpenCode/Zen specifics for the bridge:** Zen is the only agent with a first-class machine model list (`opencode models + GET /zen/v1/models`); Zen model ids use the `opencode/<model-id>` format and require an OpenCode Zen API key via `/connect`.
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## Sources verified 2026-06-10

PART A (SDK versions): - <https://github.com/aws/aws-sdk-go-v2/releases> - <https://pkg.go.dev/github.com/aws/aws-sdk-go-v2/service/bedrockruntime?tab=versions> - <https://pkg.go.dev/github.com/Azure/azure-sdk-for-go/sdk/azcore?tab=versions> - <https://pkg.go.dev/github.com/Azure/azure-sdk-for-go/sdk/azidentity?tab=versions> - <https://github.com/gin-gonic/gin/releases> - <https://github.com/smacker/go-tree-sitter/commits/master> - <https://github.com/tree-sitter/go-tree-sitter> - <https://pkg.go.dev/github.com/tree-sitter/go-tree-sitter?tab=versions> - <https://pkg.go.dev/google.golang.org/genai?tab=versions> - <https://github.com/getzep/zep-go/releases> - <https://pkg.go.dev/github.com/getzep/zep-go/v3?tab=versions> - <https://pkg.go.dev/github.com/openai/openai-go?tab=versions> - <https://pkg.go.dev/github.com/anthropics/anthropic-sdk-go?tab=versions>

PART B (CLI agents): - <https://code.claude.com/docs/en/cli-reference> (redirect from <https://docs.claude.com/en/docs/claude-code/cli-reference>) - <https://qwenlm.github.io/qwen-code-docs/en/users/features/headless/> - <https://opencode.ai/docs/cli/> - <https://opencode.ai/docs/zen/> - <https://github.com/google-gemini/gemini-cli/blob/main/docs/cli/headless.md> - <https://geminicli.com/docs/cli/headless/> (referenced) - <https://github.com/charmbracelet/crush> - <https://deepwiki.com/charmbracelet/crush/2.2-cli-usage> (referenced for `crush run` flags; [mintlify.com/charmbracelet/crush/cli/run](https://mintlify.com/charmbracelet/crush/cli/run) returned HTTP 410 Gone) - <https://developers.openai.com/codex/cli/reference> - <https://goose-docs.ai/docs/guides/goose-cli-commands/> - <https://docs.github.com/en/copilot/reference/copilot-cli-reference/cli-programmatic-reference> - <https://docs.github.com/en/copilot/how-tos/copilot-cli/automate-copilot-cli/run-cli-programmatically>

Unreachable / negative-finding pages (recorded honestly): - <https://docs.claude.com/en/docs/claude-code/cli-reference> — 301 → resolved at [code.claude.com](https://code.claude.com). - <https://qwenlm.github.io/qwen-code-docs/en/cli/>, [...](https://qwenlm.github.io/qwen-code-docs/en/cli/configuration/), [...](https://qwenlm.github.io/qwen-code-docs/en/cli/headless/) — 404; correct path is [/en/users/features/headless/](https://qwenlm.github.io/qwen-code-docs/en/users/features/headless/). - <https://github.com/google-gemini/gemini-cli/blob/main/docs/cli/configuration.md> and <https://github.com/google-gemini/gemini-cli/blob/main/docs/cli/headless.md>. - <https://www.mintlify.com/charmbracelet/crush/cli/run> — HTTP 410 Gone; Crush JSON-output absence confirmed from README + DeepWiki CLI-usage page (negative finding). - <https://block.github.io/goose/docs/guides/goose-cli-commands/> — 404; resolved via [goose-docs.ai](https://goose-docs.ai).